

### EXNO: 3

**Designing a Quarter wave transformer by varying following parameters**

**a) frequency**

**b) input impedance**

**c) characteristic impedance**

**Test Case 1:**

**Design a quarter-wave transformer for an input impedance ( $Z_{in}$ ) of 50 ohms and a load impedance ( $Z_L$ ) of 100 ohms. Analyze the transformer's performance as the operating frequency is varied across {100 MHz, 200 MHz, 300 MHz, 500 MHz}.**

**OUTPUT:**

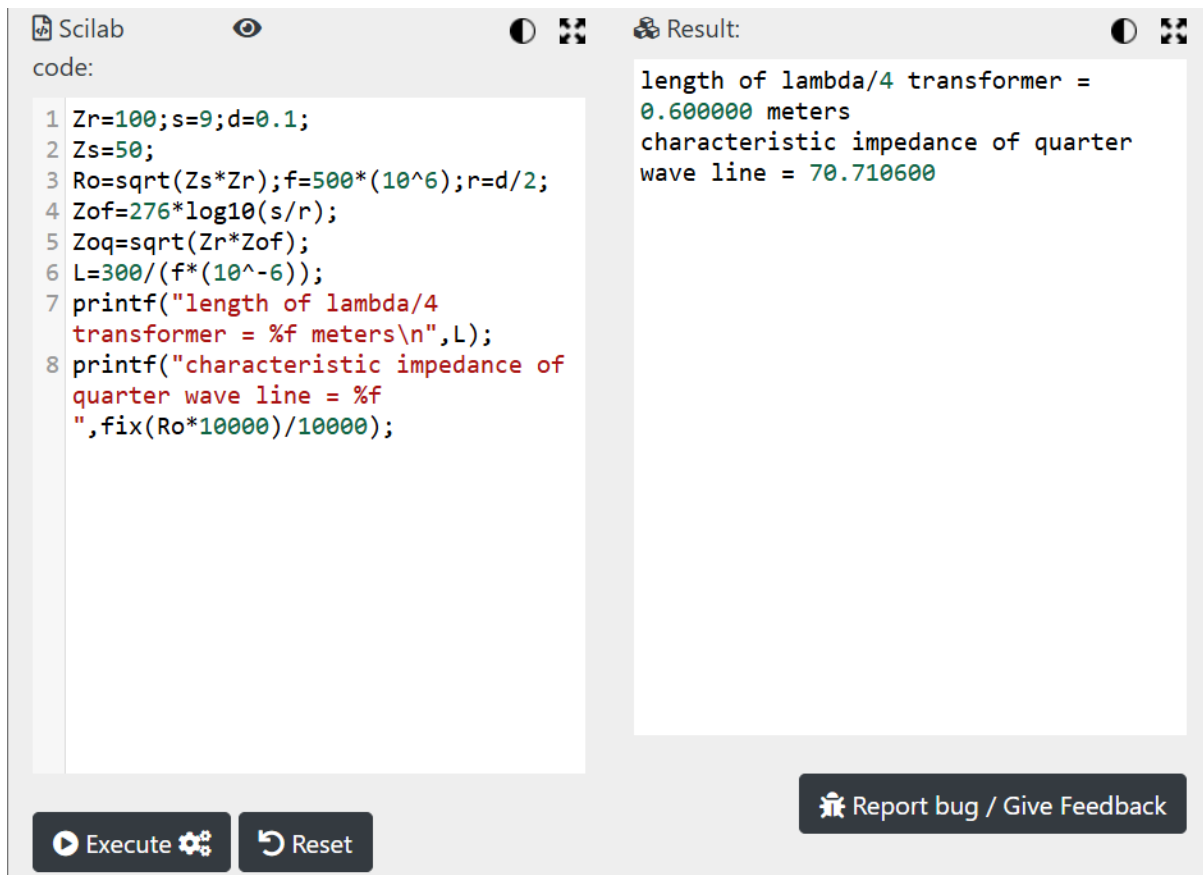
**$Z_R=100$  ohms  $Z_{in}=50$  ohms**

| <b>ZS ohms</b> | <b>ZR ohms</b> | <b>Ro ohms</b> |
|----------------|----------------|----------------|
| <b>100</b>     | <b>50</b>      | <b>70.71</b>   |

**$Z_R=100$  ohms  $Z_{in}=50$  ohms**

| <b>Frequency<br/>(MHz)</b> | <b>Length of <math>\lambda/4</math> transformer<br/>(meters)</b> |
|----------------------------|------------------------------------------------------------------|
| <b>100</b>                 | <b>0.750</b>                                                     |
| <b>200</b>                 | <b>0.375</b>                                                     |
| <b>300</b>                 | <b>0.250</b>                                                     |
| <b>500</b>                 | <b>0.150</b>                                                     |

## Simulation Results:



The image shows a Scilab simulation window. On the left, the 'code' pane contains the following Scilab script:

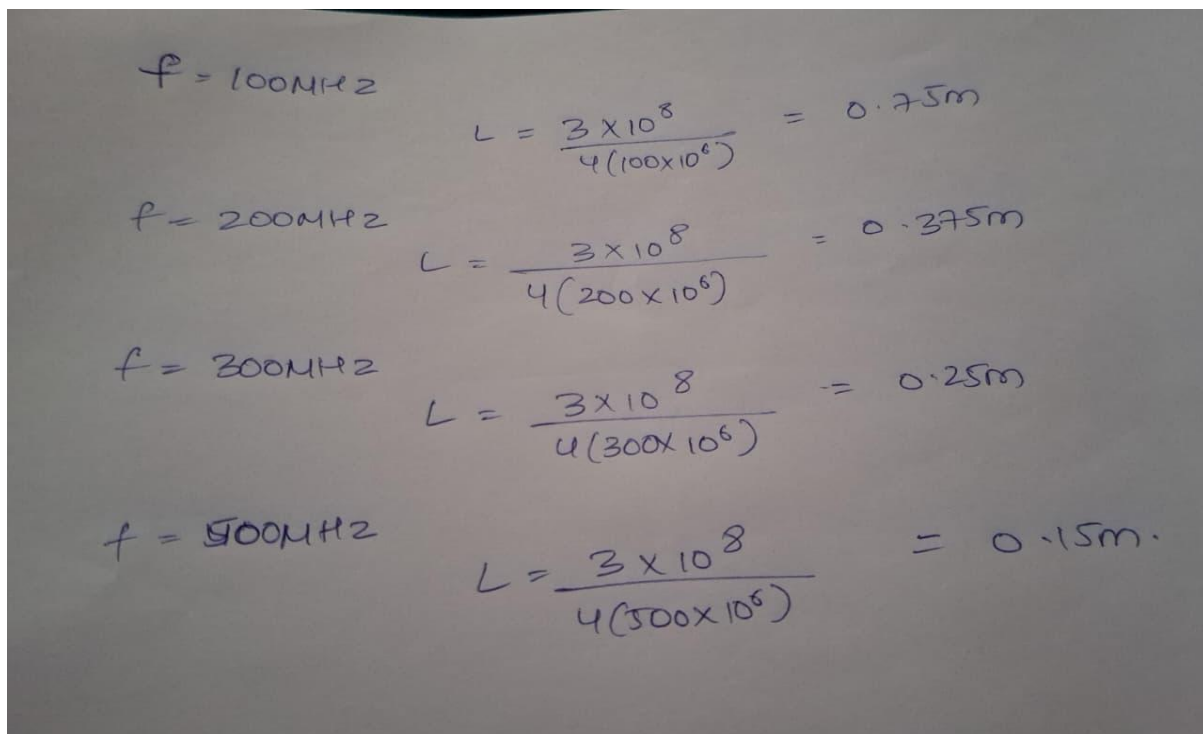
```
1 Zr=100;s=9;d=0.1;
2 Zs=50;
3 Ro=sqrt(Zs*Zr);f=500*(10^6);r=d/2;
4 Zof=276*log10(s/r);
5 Zoq=sqrt(Zr*Zof);
6 L=300/(f*(10^-6));
7 printf("length of lambda/4
transformer = %f meters\n",L);
8 printf("characteristic impedance of
quarter wave line = %f
",fix(Ro*10000)/10000);
```

On the right, the 'Result' pane displays the output of the script:

```
length of lambda/4 transformer =
0.600000 meters
characteristic impedance of quarter
wave line = 70.710600
```

At the bottom of the window, there are buttons for 'Execute', 'Reset', and 'Report bug / Give Feedback'.

## Theoretical Results



The image shows handwritten calculations for the length of a quarter-wave transformer ( $L$ ) at different frequencies ( $f$ ). The formula used is  $L = \frac{3 \times 10^8}{4(f \times 10^6)}$ .

| Frequency ( $f$ )     | Length ( $L$ )                                                   |
|-----------------------|------------------------------------------------------------------|
| $f = 100 \text{ MHz}$ | $L = \frac{3 \times 10^8}{4(100 \times 10^6)} = 0.75 \text{ m}$  |
| $f = 200 \text{ MHz}$ | $L = \frac{3 \times 10^8}{4(200 \times 10^6)} = 0.375 \text{ m}$ |
| $f = 300 \text{ MHz}$ | $L = \frac{3 \times 10^8}{4(300 \times 10^6)} = 0.25 \text{ m}$  |
| $f = 500 \text{ MHz}$ | $L = \frac{3 \times 10^8}{4(500 \times 10^6)} = 0.15 \text{ m}$  |

## Test Case 2:

For a fixed operating frequency of 2 GHz and a load impedance (ZL) of 100 ohms, design a quarter-wave transformer and evaluate its performance as the input impedance (Zin) varies through {25 ohms, 50 ohms, 75 ohms, 100 ohms}. Validate that the transformer's characteristic impedance ( $Z_{\text{transformer}} = R_o = \sqrt{Z_{\text{in}} \cdot Z_L}$ ) changes with Zin.

## OUTPUT:

ZR=100 ohms Variable: Zin

| Zin | Ro     |
|-----|--------|
| 25  | 50     |
| 50  | 70.71  |
| 75  | 86.60  |
| 100 | 100.00 |

## Simulation Results:

The image shows a Scilab simulation window with a code editor on the left and a result window on the right. The code defines parameters for a quarter-wave transformer and calculates its length and characteristic impedance. The results window displays the calculated values.

```
Scilab
code:
1 Zr=100;s=9;d=0.1;
2 Zs=100;
3 Ro=sqrt(Zs*Zr);f=500*(10^6);r=d/2;
4 Zof=276*log10(s/r);
5 Zoq=sqrt(Zr*Zof);
6 L=300/(f*(10^-6));
7 printf("length of lambda/4
transformer = %f meters\n",L);
8 printf("characteristic impedance of
quarter wave line = %f
",fix(Ro*10000)/10000);

Result:
length of lambda/4 transformer =
0.600000 meters
characteristic impedance of quarter
wave line = 100.000000
```

Execute Reset [Report bug / Give Feedback](#)